

Udemy Course
ML, DL, NLP
Krushnak Sir

Sec-22

Inferential
Statistics

Lec-5 : T Stats with
T test hypothesis

Lec-6 2 test vs
T test

Lec-7 type I & type II
Error

T-stats :- T test
→ one sample
t-test

- ① In the population, the average IQ is 100. A team of researchers want to test a new medication to see if it has a positive or negative effect on intelligence or not.

A sample of 30 participants were given the medication and the mean IQ was 140 with SD deviation of 20.

Did the medication affect intelligence?

Sol

The total mean = 100.

No. of participants = 30

α (confidence interval) = 95%

$$\therefore \alpha = 1 - 0.95 \\ = 0.05$$

$$\mu = 100 \quad n = 30 \\ \bar{x} = 140 \quad s = 20$$

$$CI = 95\% \quad \alpha = 0.05$$

Null hypothesis $H_0 = \mu = 100$

(3)

Alternative Hypotheses

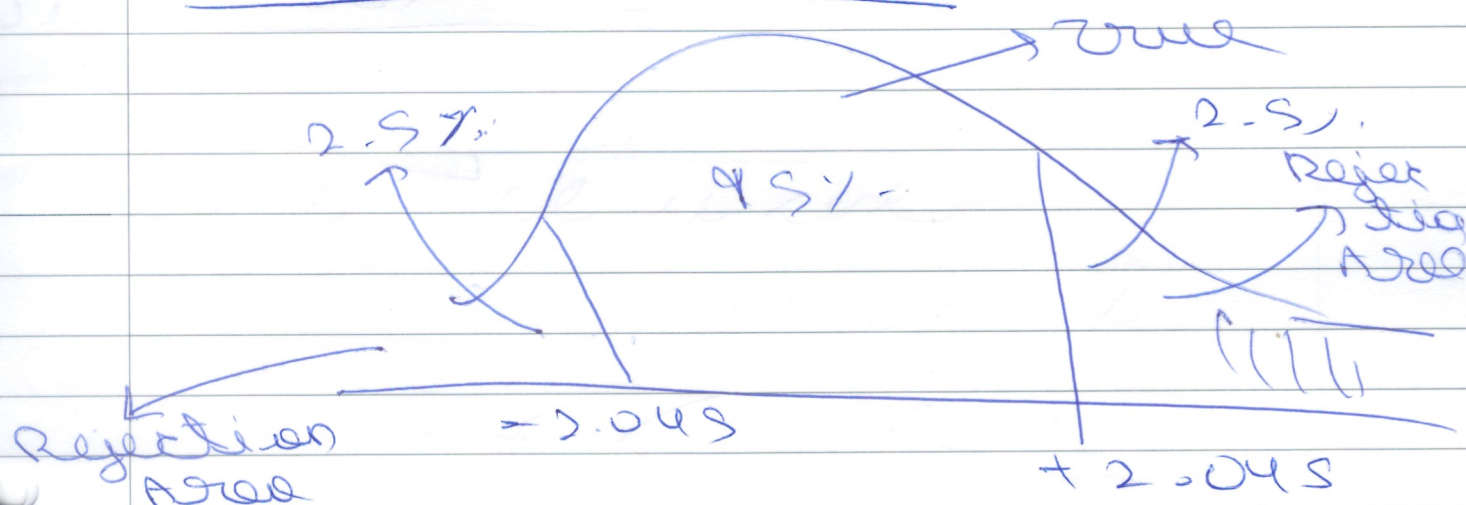
$$H_1 = \mu \neq 100$$

(2) $\alpha = 0.05$

(3) Degree of Freedom

$$df = n - 1 = 30 - 1 = 29$$

(4) Decision Rule



if t -test is less than -2.049 or greater than 2.049 reject the null hypothesis

⑤

Calculate the test statistics

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Formula

$$= \frac{140 - 100}{\frac{20}{\sqrt{30}}}$$

$$= \frac{40}{3.08} = 10.96$$

This is the t-test value

$$t = 10.96$$

$$10.96 > 2.045$$

→ it falls in the rejection region

Since $t = 10.96 > 2.045$

→ reject the Null Hypothesis.

Hence medication used has affected IQ.

it has basically increased the intelligence.

100-6

2 test vs T test

④ How to use 2 test vs T test

→ this depends on our knowledge of population standard deviation

Do you know the
population std.
deviation

yes

no

Is the
sample
size above
30?

Use the
t-test

yes

no

Use the
z test

Use the
t-test

lec-7

Type 1 and Type 2 Error

Reality :-

Null Hypothesis is True,
or Null Hypothesis
is False.

Decision :-

Null Hypothesis is true
or the Null Hypothesis
is False.

Outcome :-

We reject the Null
Hypothesis in reality
when it is false
→ Good

Outcome 2:-

We reject the Null Hypothesis when in reality, it is the ~~reality~~ true.

→ Type I Error

Outcome 3:-

We retain the Null Hypothesis, when in reality it is False

→ We retain the Null Hypothesis, when in reality it is False

→ Type 2 Error

Outcome 4:-

We retain the Null Hypothesis when in reality it is true
→ Good